

The edge contractive lift

Euler tail, Woodbury Δ , and the r401 chart: what the edge half of Satz (C) actually is

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Abstract

After rounds 400–403 the bulk of $R_{N-3} - \frac{1}{2}I$ has a single defect candidate and the edge is an exact 3×3 normal form $\Phi_{\text{edge}}(a, b)$ with living/dead coordinate sch . The reviewer (DCCLXIX) asks that the missing Euler null mode be written $\ell = Vc$ from geometric recurrences, that $1 - \|c\|^2$ be a manifest positive source sum, and that this reserve equal $\kappa(a, b)(-\text{sch})$ in the r401 Sylvester chart, with dead χ appearing as $\|c\| > 1$. The Euler tail, the disk Parseval identity, and the Woodbury identity $\Delta = \kappa_{\text{closed}}(-\text{sch})$ are finite SATZ. The source-explicit lift of the *constant* on Y has residual 0. The Sherman–Morrison border is not the aggregated tail $z_p^{K_p+1}$; dead χ keep $\|c\| < 1$ (death remains $\text{sch} > 0$); κ is not a function of (a, b) alone; the geometric sum and Woodbury Δ differ by two to four orders of magnitude. The A_0 -defect lift stays open (round 404: class 3 not reached; $|\cos|(v_-, 1) = 0.007$). No RH claim.

Status. Lemmas 1–3 are proved (finite identities). Propositions 1–3 are finite machine checks. No RH claim. No L^* claim. No $R^\dagger \succ \frac{1}{2}I$ claim.

1 What is asked

Write $A_0 = R_{N-3} - \frac{1}{2}I$ on the hole nodes Y and $M^\dagger = A^\dagger + UJU^T$ for the mixed form of round 369, with $J = \text{diag}(1, 1, 1/\text{den})$ and $\Phi = J^{-1} + U^T(A^\dagger)^{-1}U$ the r401 3×3 . If a source-pure $H \succ 0$ and a source-pure ℓ satisfy $A = H - \ell\ell^T$ on the same space as U , Woodbury gives

$$\Delta = 1 - \eta + r^T K_W^{-1} r, \quad \eta = \ell^T H^{-1} \ell, \quad r = U^T H^{-1} \ell, \quad K_W = J^{-1} + U^T H^{-1} U, \quad (1)$$

and $M = H + UJU^T - \ell\ell^T \succ 0$ if and only if $\Delta > 0$. The constructive form is $\ell = Vc$ with $V = [F_\Lambda, F_\Gamma, F_{U_V}, UJ^{1/2}]$ and $\|c\| < 1$. The Euler tail

$$1 - z_p^{K_p+1} = (1 - z_p) \sum_{r=0}^{K_p} z_p^r, \quad z_p = p^{-1/2} e^{i\vartheta_p}, \quad (2)$$

is the proposed source of c , with $1 - \|c\|^2$ equal to

$$\eta_\Gamma + \eta_{\text{pole}} + \sum_p (1 - p^{-1}) |b_p|^2 + \sum_p p^{-(K_p+1)} |t_p|^2 \quad (3)$$

and, in the r401 chart, to $\kappa(a, b)(-\text{sch})$.

2 Finite identities

Lemma 1 (Euler tail). *For every z and every integer $K \geq 0$, $1 - z^{K+1} = (1 - z) \sum_{r=0}^K z^r$. Over $\mathbb{Q}$, at $z = 1/3$ and $K = 5$, both sides equal $728/729$. Omitting the tail leaves the remainder $z^{K+1} \neq 0$.*

Lemma 2 (disk Parseval). *For $0 < q < 1$ and $K \geq 0$, $(1 - q) \sum_{r=0}^K q^r = 1 - q^{K+1}$. At $q = 1/2$ and $K = 6$ the left side is $127/128$ and the reserve q^{K+1} is $1/128$. With $q = p^{-1} = |z_p|^2$ this is (3) for a single prime ($b_p = 1$, no Gamma/pole).*

Lemma 3 (Woodbury / Schur). *Let $H \succ 0$ (or merely invertible), $A = H - \ell \ell^T$ invertible, and $\Phi = J^{-1} + U^T A^{-1} U$. Write sch for the scalar Schur complement of the leading 2×2 of Φ , and K_2 for that block. Then*

$$\Delta = (1 - \eta) \frac{\det \Phi}{\det K_W} = \kappa_{\text{closed}}(-\text{sch}), \quad \kappa_{\text{closed}} = -(1 - \eta) \frac{\det K_2}{\det K_W}.$$

Over $\mathbb{Q}$, on $H = \text{diag}(2, 3, 4)$, $\ell = (1, 1, 1)$, $U = I$, $J^{-1} = \text{diag}(1, 1, 2)$: $\eta = 13/12$, $\Delta = 7/36$, $\text{sch} = 21/8$, $\kappa_{\text{closed}} = -2/27$, and the identity is exact. After the r401 Sylvester map D , $\det K_2 = \det J_2 / \det(D_2)^2$, so κ_{closed} still depends on the bulk split (H, ℓ) through η and K_W : it is not a function of the chart coordinates (a, b) alone.

Lemma 4 (Cauchy Gram). *The 2×2 Cauchy Gram $G_{ij} = 1/(z_i z_j)$ at $z = (1, 2)$ has $\det G = 1/72 > 0$. This is the F_Γ block over $\mathbb{Q}$; the lift below takes its coefficients to be zero.*

3 The geometric lift of the constant

On Y set $\vartheta_j = \arccos(y_j)$ and $z_p(j) = p^{-1/2} e^{i\vartheta_j \log p}$. Let $\nu_p = \log p / \sum \log p$ over a finite prime list and $E_p = (1 - z_p) \sum_{r=0}^K z_p^r$, $T_p = z_p^{K+1}$. Then (2) gives $E_p + T_p = 1$ pointwise, hence

$$\mathbf{1} = \sum_p \nu_p (E_p + T_p) \tag{4}$$

with explicit c (the weights ν_p on the real/imaginary parts of E_p and T_p). No optimiser.

Proposition 1 (ones-lift). *On FRAME-A, $\|\mathbf{1} - \sum \nu(E + T)\|_2 \leq 10^{-12}$. Omitting T leaves residual 0.049. The same identity continues to hold after a 10^{-4} jitter of the nodes (it is an identity in z). The bulk defect v_- of A_0 is not this vector: $|\cos|(v_-, \mathbf{1}) = 0.007$ and the residual of v_- against the geometric V is 0.81. Round 404 independently records that the ones-mode is not the null mode of A_0 .*

Proposition 2 (border is not the tail). *The Y -component s of the Sherman–Morrison border of round 401 has relative residual 0.996 against $\text{span}\{\text{Re } T, \text{Im } T\}$ and correlation 0.07 with $\text{Re } T$, on the named phase $\vartheta_j \log p$. Every other source-explicit phase convention tested (shared $\arccos y$, $y \log p$, Chebyshev T_5) leaves relative residual ≥ 0.96 . The identification “border = aggregated z^{K+1} ” is refuted for these conventions.*

4 Woodbury Δ of the ones-split, and worlds

The mixed $A^\dagger = \text{blkdiag}(A_0, -1/2)$ is not a rank-one update of a positive H (two negatives). The ones-split used for measurement is the Y -only object $H = A_0 + \mathbf{1}\mathbf{1}^T$ (normalised) with the two CD columns, which is the bulk half of (1). Then Δ and $\|c_{\min}\|^2 = \ell^T B^{-1} \ell = 1 - \Delta$ are closed forms, not an optimisation.

Proposition 3 (ones-split census). *On FRAME-A, $\Delta = 2.110 \cdot 10^{-3}$, $\|c\|^2 = 0.99789$, $\kappa = \Delta/(-\text{sch}) = 0.0315$ with $\text{sch} = -0.06696$. On core-42: $\Delta > 0$, $\kappa > 0$, and $0 < \|c\|^2 < 1$ on 42/42, with $\Delta \in [6.7 \cdot 10^{-6}, 2.11 \cdot 10^{-3}]$. Living χ_3 : 37/37 have $\kappa > 0$ and $\|c\| < 1$. Dead χ_3 : 5/5 have $\text{sch} > 0$, $\kappa < 0$, and $\|c\| < 1$. The predicted overflow $\|c\| > 1$ on dead χ is refuted for this source c . Death remains $\text{sch} > 0$ (round 401). The geometric disk sum (3) is $O(10^{-1})$ on the named primes; Woodbury Δ is $O(10^{-5}-10^{-3})$: they are not the same scalar, so “sum form = $\kappa(-\text{sch})$ ” is refuted on scale. Weight permutation already yields $\text{ind}_- A_0 \geq 15$ (round 403): factorisation (G) breaks before any lift. Scramble: $\text{ind}_- = 21$. Wrong-sign J : mixed residual 867/115 $\neq 0$.*

5 What this is not

This is not a proof of $R^\dagger \succ \frac{1}{2}I$, not a source-explicit $A_0 = H - \ell\ell^T$ (round 404 stopped at class 3), and not a unique $\kappa(a, b)$ in the r401 chart. The finite SATZ are the tail, the disk identity, and the Woodbury–Schur determinant formula. The A_0 -defect lift, and the identification of the mixed border with an Euler tail, remain open or refuted as named. No RH claim.

References

- [1] S. Hamann, *The augmented edge signature*, `rh/problem/edge_signature.tex`, 2026.
- [2] S. Hamann, *Source one-defect Gram*, `rh/problem/one_defect_gram.tex`, 2026.